Rb Beam Flux Measurement via Absorption Spectroscopy: Derivation, Assumptions, and Validity

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1. Overview of the Measurement

The total Rb beam flux (atoms/s) is determined by combining two separate measurements: (i) an absolute absorption measurement at the beam center, which gives the column density of Rb atoms along the probe beam path, and (ii) a divergence scan, in which the probe beam is translated vertically across the atomic beam profile, giving the relative spatial distribution of atoms. The divergence profile is rescaled to absolute column density using the center measurement, integrated across the beam, and multiplied by the mean thermal velocity to obtain the flux.

The chain of reasoning is:

Fractional absorption → Optical density → ^{85}Rb F=3 column density → Total Rb column density → Rescaled beam profile → Integrated line density → Flux = (line density) x (mean speed)

2. Beer-Lambert Law and Column Density

2.1 The Law

When a collimated, monochromatic probe beam of intensity I_0 passes through an absorbing medium of length L, the transmitted intensity I is given by the Beer-Lambert law [1, 2]:

$$I = I_0 \exp(-OD)$$

where OD is the optical density (also called absorbance in the natural-log convention). For an atomic vapor with number density n(x) varying along the probe path x, the optical density is:

$$OD = \sigma_{eff} * \int n(x)\, dx = \sigma_{eff} * N_{col}$$

where σ_{eff} is the effective resonant absorption cross section for the transition being driven, and $N_{col} = \int n(x)\, dx$ is the column density (atoms/cm^2) — the number of atoms per unit area integrated along the probe beam path. The key point is that absorption directly measures the path-integrated column density, not the local volume density.

2.2 Extracting Column Density from Measured Absorption

If the measured fractional absorption is A (in percent), the transmission is T = 1 - A/100, and the optical density is OD = -ln(T). The column density of atoms in the specific quantum state

being probed is then:

$$N_{col,state} = OD / sigma_{eff}$$

For the ^{85}Rb D_2 F=3 -> F'=4 transition, $sigma_{eff}$ = 1.248 x 10^{-9} cm^2 [3]. This is the cross section averaged over all m_F sublevels assuming uniform (thermal) population, and is valid for any probe polarization in this limit (see Section 5).

2.3 Converting to Total Rb Column Density

The absorption measurement probes only ^{85}Rb atoms in the F=3 ground state. To recover the total Rb column density, two correction factors are applied:

(a) Hyperfine population fraction $f_{F=3}$: At thermal equilibrium with kT >> hyperfine splitting (~3 GHz for ^{85}Rb ground state, while kT/h ~ 10 THz at 500 K), the population distributes among sublevels proportional to their degeneracy (2F+1). For ^{85}Rb with I=5/2, the ground state has F=2 (5 sublevels) and F=3 (7 sublevels), so:

$$f_{F=3} = (2*3+1) / [(2*2+1) + (2*3+1)] = 7/12$$

(b) Isotopic abundance: Natural rubidium is 72.17% ^{85}Rb and 27.83% ^{87}Rb [3]. Since only ^{85}Rb is probed, dividing by P_{85} = 0.7217 gives the total Rb column density:

$$N_{Rb} = OD / (sigma_{eff} * f_{F=3} * P_{85})$$

3. The Linearity Assumption: When Is Absorption Proportional to Atom Number?

A central assumption of this measurement is that the relative peak heights in the divergence scan are proportional to the column density at each probe position — i.e., a peak half as tall means half as many atoms. This is a statement about the Beer-Lambert law operating in the optically thin regime.

3.1 The Optically Thin Limit

The fractional absorption is A = 1 - exp(-OD). For small OD (low column density):

$$A \approx OD = sigma_{eff} * N_{col} \quad (for\ OD << 1)$$

In this regime, absorption is linear in atom number. For large OD, the exponential compresses the response: doubling the atoms does **not** double the absorption. For example, at OD = 0.1 the error from the linear approximation is ~5%, while at OD = 0.5 it grows to ~20%.

3.2 Checking Your Data

To determine whether your data is in the linear regime, estimate the optical density at the beam center from your absolute absorption measurement. If A is your peak absorption in percent:

A (%)	OD = -ln(1-A/100)	Linear approx. error
1%	0.010	< 1% (excellent)
5%	0.051	~ 2.5% (good)
10%	0.105	~ 5% (acceptable)
20%	0.223	~ 11% (caution)
50%	0.693	~ 31% (nonlinear)

If your measured absorption is below ~10%, you are safely in the linear regime and the proportionality assumption holds well. For higher absorptions, using OD = -ln(T) instead of the fractional absorption A directly restores linearity in N_{col}, which is exactly what is done in the flux code.

3.3 Other Conditions for Beer-Lambert Validity

Beyond the optically thin condition, Beer-Lambert requires [1, 4, 5]: (a) the probe must not saturate the transition ($I << I_{sat}$), avoiding stimulated emission and ground-state depletion; (b) no significant optical pumping occurs during the atom's transit through the probe beam, so the sublevel populations remain thermal; (c) the probe is monochromatic relative to the absorption linewidth (true for a single-mode ECDL scanning across the line); and (d) the medium is not scattering (true for a dilute atomic beam in vacuum).

For your setup with ~1 uW probe power and a ~0.5 mm beam diameter, I ~ 0.05 mW/cm^2, which is well below I_{sat} = 3.90 mW/cm^2 (isotropic, F=3->F'=4 [3]). The weak-probe condition is satisfied. The transit time through the 0.5 mm beam at the most probable thermal speed (~350 m/s at 500 K) is ~1.4 us, much shorter than the optical pumping time t_{pump} ~ (Gamma * s)$^{-1}$ ~ 0.5 ms at this intensity. So optical pumping is negligible and Beer-Lambert applies [5].

4. From Divergence Profile to Flux

4.1 Rescaling the Divergence Profile

The divergence scan measures relative absorption as a function of probe beam height y. Call these values $a(y_i)$ (in arbitrary units, since the amplification chain is not calibrated to absolute absorption). At the beam center position y_0, a separate absolute absorption measurement gives the true column density $N_{Rb}(y_0)$. The rescaling factor is:

$$R = N_{Rb}(y_0) / a(y_0)$$

The absolute column density profile is then $N_{Rb}(y) = R * a(y)$. This assumes the amplification is the same for all measurements in the divergence scan, which is ensured by keeping the electronics and probe power fixed. A Gaussian fit to the divergence data provides a smooth, analytic profile for integration.

4.2 Integration Across the Beam

The column density $N_{Rb}(y)$ has units of cm^{-2}. It is the integral of the volume density n(x,y) along the probe beam path x. Integrating over the transverse direction y gives the line density:

$$\text{Lambda} = \text{integral of } N_{Rb}(y)\, dy \text{ [units: } cm^{-1}\text{]}$$

This is the total number of Rb atoms per unit length along the beam propagation direction z, summed over the full beam cross section. Note that no Abel inversion is required: the x-integration is performed physically by the probe beam, and the y-integration is performed numerically. The result Lambda captures the full 2D cross section of the beam.

4.3 From Line Density to Flux

Consider a thin slab of thickness dz perpendicular to the beam. The number of atoms in this slab is dN = Lambda * dz. Each atom traverses dz in time dt = dz/v, where v is its axial velocity. The rate atoms pass through the slab is:

$$\text{Phi} = dN/dt = \text{Lambda} * dz / (dz / v) = \text{Lambda} * v \text{ [atoms/s]}$$

The probe beam thickness dz cancels exactly, which is why it does not appear in the final expression. This is a standard result for relating density to flux [6].

4.4 Which Velocity to Use

For an effusive beam from an oven at temperature T, the appropriate velocity for flux calculations is the **mean speed** of the Maxwell-Boltzmann distribution [6, 7]:

$$v_{mean} = \text{sqrt}(8\, k_B\, T / (\text{pi} * m))$$

This is the first moment of the speed distribution, which is the relevant average when computing flux = density x velocity. Note that this differs from the most probable speed v_{mp} =

sqrt(2 k_B T / m) by a factor of sqrt(4/pi) ~ 1.13. At T = 503 K (oven at 230 C) and m = 85.47 u:

Quantity	Formula	Value (m/s)	Value (cm/s)
Most probable speed	sqrt(2kT/m)	310	3.10 x 10^4
Mean speed	sqrt(8kT/(pi*m))	350	3.50 x 10^4
RMS speed	sqrt(3kT/m)	380	3.80 x 10^4

The original code used v_{mp} = sqrt(2kT/m); correcting to v_{mean} increases the computed flux by ~13%.

5. Absorption Cross Section and Polarization Independence

For the ^{85}Rb D_2 F=3 -> F'=4 transition with thermally populated (uniform) m_F sublevels, the effective cross section averaged over all sublevels is independent of the probe polarization. This can be verified by summing the squared Clebsch-Gordan coefficients from the Steck data tables [3]:

Polarization q	Sum of |CG|^2 over m_F	Average per sublevel
sigma+ (q=+1)	84/56 = 3/2	3/14
pi (q=0)	84/56 = 3/2	3/14
sigma- (q=-1)	84/56 = 3/2	3/14

All three polarizations give identical sums. The effective cross section is therefore σ_{eff} = (3/14) * $\sigma_{0,cycling}$ = (3/14) * 3*lambda^2/(2*pi) = 1.248 x 10^{-9} cm^2, with corresponding I_{sat} = 3.895 mW/cm^2. This is a consequence of the Wigner-Eckart theorem (Steck Eq. 41) [3]. Polarization **does** matter when optical pumping redistributes the m_F populations (e.g. circularly polarized light pumps atoms to the stretched state), but this requires sufficient interaction time, which is not the case for a weak probe on a fast thermal beam.

6. Complete Flux Formula

Combining all the steps, the total Rb beam flux is:

Step 1: From measured absorption A (percent) at beam center:

$$T = 1 - A/100$$

$$OD = -\ln(T)$$

$$N_{Rb} = OD / (\sigma_{eff} * f_{F=3} * P_{85})$$

Step 2: Rescale divergence Gaussian fit to absolute units:

$$N_{Rb}(y) = [N_{Rb} / a(y_0)] * G(y; A, y_0, \sigma)$$

Step 3: Integrate across beam:

$$\Lambda = \text{integral of } N_{Rb}(y)\, dy \; [cm^{-1}]$$

Step 4: Multiply by mean speed:

$$\Phi = \Lambda * \sqrt{8 k_B T / (\pi * m)} \; [atoms/s]$$

7. Summary of Constants Used

Quantity	Symbol	Value	Source
Absorption cross section (iso, F=3->F'=4)	sigma_eff	1.248 x 10^-9 cm^2	Steck [3]
Saturation intensity (iso, F=3->F'=4)	I_sat	3.895 mW/cm^2	Steck [3]
F=3 population fraction	f_F=3	7/12 = 0.5833	Thermal equil.
85Rb abundance	P_85	0.7217	Steck [3]
85Rb mass	m	85.47 u	Steck [3]
Boltzmann constant	k_B	1.381 x 10^-23 J/K	CODATA

8. Key Assumptions and Caveats

(1) Weak probe / no optical pumping: The probe intensity must be well below I_{sat} so that stimulated emission is negligible and the m_F sublevel populations remain thermal during transit. At I ~ 0.05 mW/cm^2 this is easily satisfied [5].

(2) Linearity (optically thin): The divergence profile uses relative peak heights as proxies for relative column densities. This is strictly valid when OD << 1. If the peak absorption at center is > ~10%, a nonlinearity correction should be considered, or equivalently, one should use OD = -ln(1 - A/100) at each scan position rather than A itself.

(3) Gaussian beam profile: The divergence profile is fit to a Gaussian. This is expected for the far-field profile of an effusive source with a channel array [7], but deviations (e.g. from misaligned capillaries or beam clipping) would introduce integration error.

(4) Single velocity: The flux formula uses a single mean speed. In reality there is a distribution of velocities. For an effusive source, the flux-weighted distribution is P(v) proportional to $v^3 \exp(-mv^2/(2kT))$, whose mean is sqrt(9*pi*kT/(8m)) [6]. Using the bulk mean speed sqrt(8kT/(pi*m)) introduces a ~5-10% systematic depending on the exact velocity weighting appropriate for your geometry. This is a second-order correction.

(5) 1D scan vs 2D cross section: The measurement scans in y and integrates over x (via absorption). If the beam is cylindrically symmetric about z, this captures the full cross section without needing an Abel inversion. If the beam is asymmetric in x vs y (e.g. from nozzle geometry), a second scan in x would be needed.

References

[1] Beer-Lambert law. See, e.g., Corney, *Atomic and Laser Spectroscopy* (Oxford, 1977), Chap. 8; or Demtroder, *Laser Spectroscopy*, 4th ed., Vol. 1, Chap. 1.

[2] Siddons, Adams, Ge, and Hughes, "Absolute absorption on rubidium D lines: comparison between theory and experiment," J. Phys. B **41**, 155004 (2008). arXiv:0805.1139.

[3] Steck, "Rubidium 85 D Line Data," revision 2.3.4 (2025). Available at http://steck.us/alkalidata. See Table 7 for cross sections / saturation intensities, Table 8 for transition strength factors S_FF', and Tables 9-11 for individual m_F Clebsch-Gordan coefficients.

[4] Sherlock and Hughes, "How weak is a weak probe in laser spectroscopy?," Am. J. Phys. **77**, 111-115 (2009). Analyzes the transit-time condition for weak-probe validity in Rb vapor.

[5] Wikipedia, "Beer-Lambert law," conditions for validity: monochromatic, collimated, non-saturating probe; no optical pumping; homogeneous, non-scattering medium. Also see Ref. [4].

[6] Ramsey, *Molecular Beams* (Oxford, 1956), Chap. 1. Standard treatment of effusive beam velocity distributions, flux = n * v_mean, and the distinction between density-weighted and flux-weighted speed distributions.

[7] Senaratne, Rajagopal, Geiger, Fujiwara, Lebedev, and Weld, "Effusive atomic oven nozzle design using an aligned microcapillary array," Rev. Sci. Instrum. **86**, 023105 (2015). Demonstrates microcapillary beam source characterization using absorption spectroscopy.

[8] Metcalf and van der Straten, *Laser Cooling and Trapping* (Springer, 1999). See Chap. 2 for atomic beam properties and velocity distributions.